

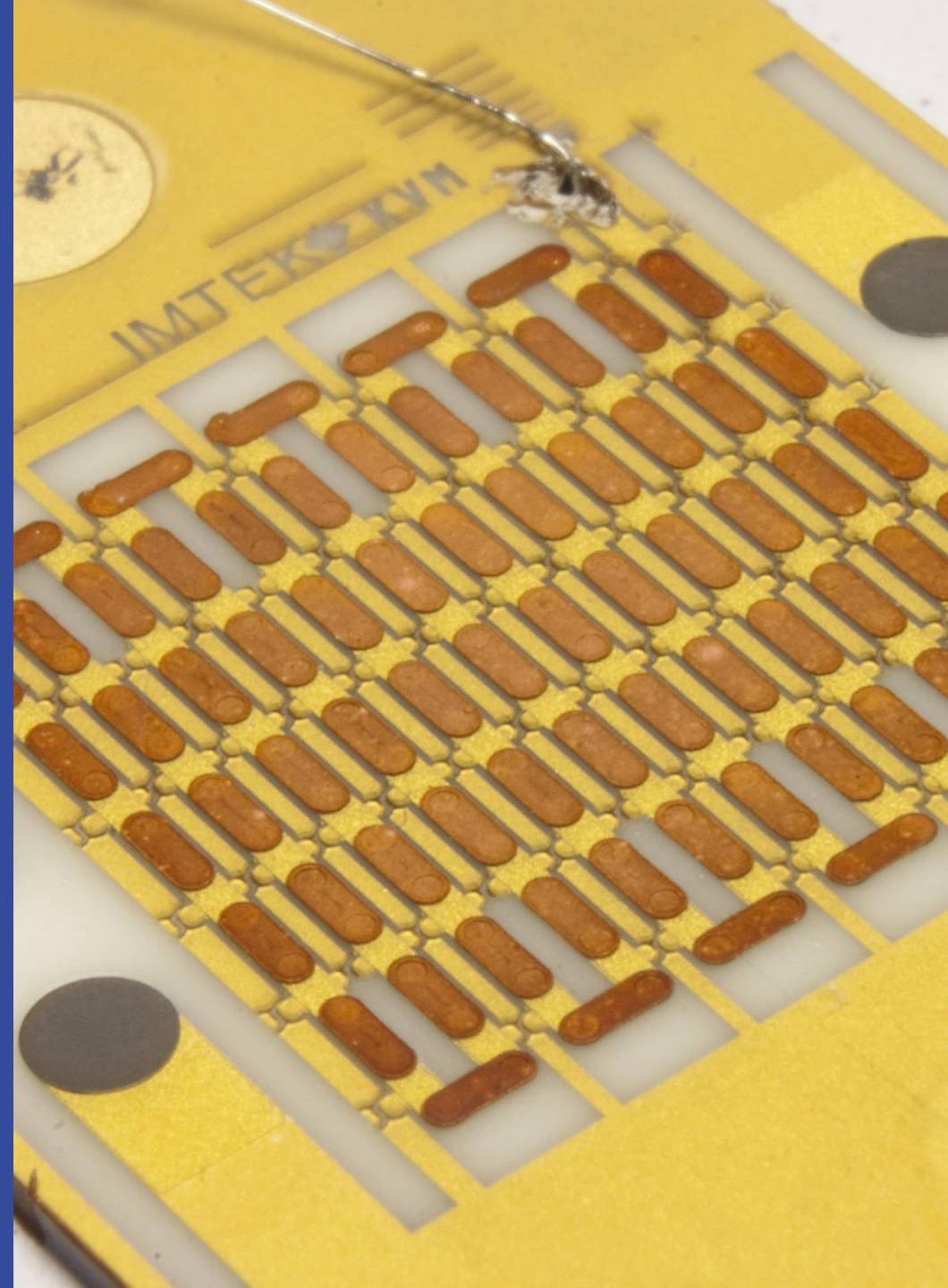
Micro Energy Harvesting

Chapter 8: Storage Concepts II - Rechargeable Batteries

Micro Energy Harvesting SoSe 2026

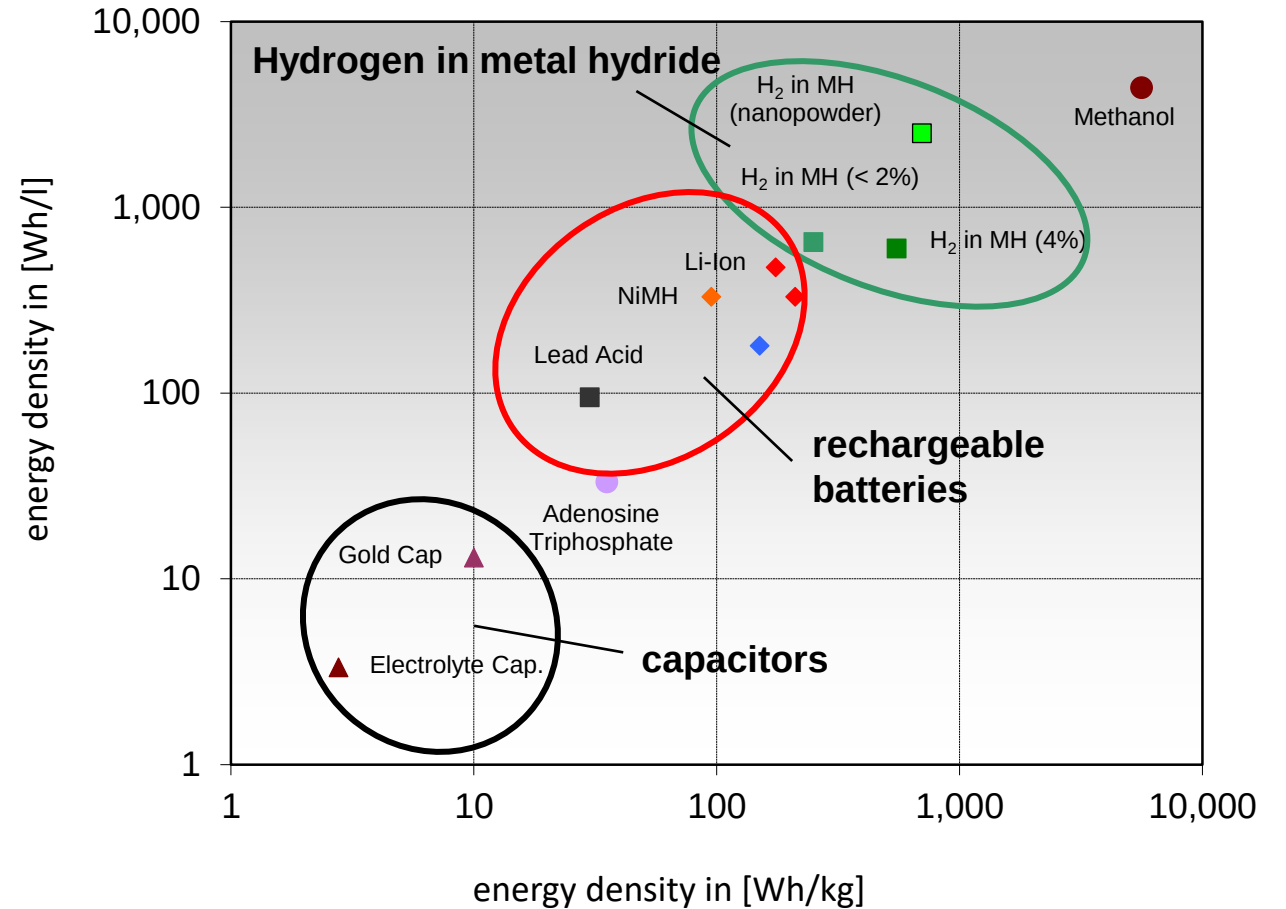
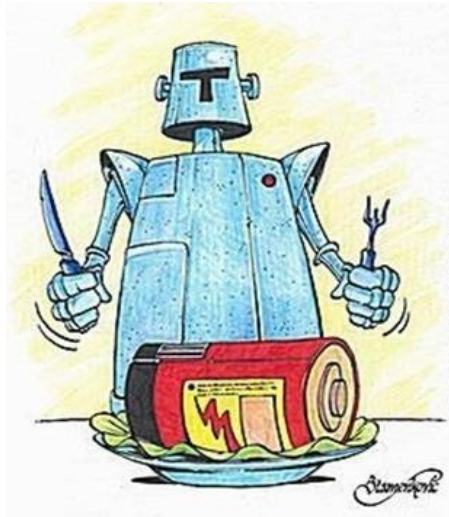
IMTEK – Lehrstuhl für Konstruktion von Mikrosystemen

Prof. Dr. Peter Woias, Dr. Suman Kundu, Dr. Frank
Goldschmidtboing



Energy Storage for Embedded Systems

- today mostly electric
- slowly emerging alternatives (e.g. thermal)
- design conflict between ...
 - storage capacity,
 - size/weight and
 - lifetime
 - ...



Refs: J. Brodd et al, *J. Electrochem. Soc.*, 151 (3), 2004, K1-K11 and HERA Hydrogen Storage Solutions, D

Energy Storage for Embedded Systems

Global Properties

Capacitors

- dielectric capacitors
 - dielectric multilayer capacitors (MLCC)
 - Tantal capacitors
 - electrolyte capacitors
 - Gold caps ...
- **charge storage**
 - (mostly) linear voltage-charge characteristics: $Q = C \cdot U$
 - dischargeable to 0 V
 - internal leakage currents ➡ electrical self-discharge
 - minimum charge current to overcome leakage effects
 - requires an overvoltage protection

Rechargeable batteries

- Pb
 - NiCd, NiMH
 - Li-Ion
 - Li-Po
- **chemical energy storage**
 - non-linear voltage-charge characteristics
 - usually not dischargeable to 0 V
 - chemical and electrical self-discharge
 - requires a dedicated charge and discharge management

Rechargeable Batteries

Overview

	Ni-Cd	Ni-MH	Pb-Acid	Li-ion		
				Co (LCO)	Mn (LMO)	PO ₄ (LFP)
Gravimetric energy density [Wh/kg]	45...80	60...120	30...50	150...250	100...150	90...120
Cycle life (down to 80% initial capacity)	1000	300...500	200...300	500...1000	500...1000	1000-2000
Self-discharge rate [1/month]	20 %	30 %	5 %	< 5 %		
Nominal cell voltage	1.25 V	1.25 V	2 V	3.6 V	3.7 V	3.2 - 3.3 V
Fast charge time	1 hr	2...4 hr	8...16 hr	1.5...3 hr	1 hr or less	
Overcharge tolerance	moderate	low	high	very low, no trickle charge		
Temperature at discharge [°C]	-40...60°C	-20...60°C	-20...60°C	-20...60°C		
Maintenance (e.g. recharge, etc.)	30...60 days	60...90 days	3...6 months	not required		
Toxicity	highly toxic	rel. low toxicity	toxic	low toxicity		

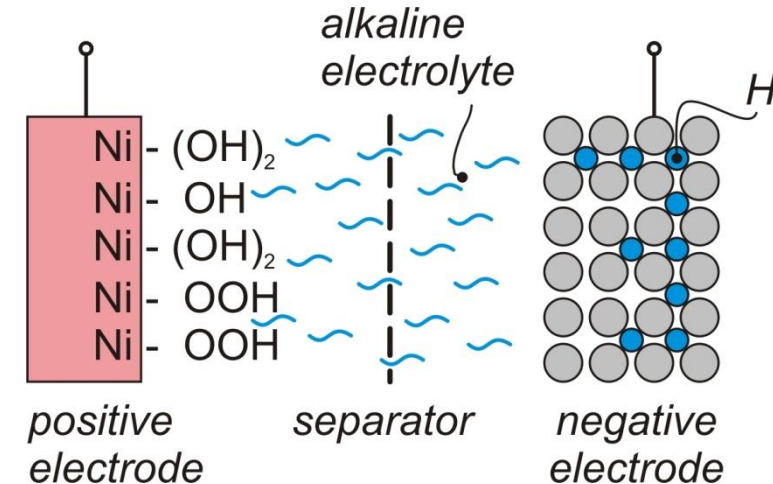
Nickel-Metal Hydride (NiMH) Batteries

Set-up and Working Principle

- use of hydrogen for energy storage
- generation of H^+ and OH^- via electrolysis
 - ➔ *negative electrode: absorption of H*
 - ➔ *positive electrode: consumption of OH^- via oxidation*

Materials

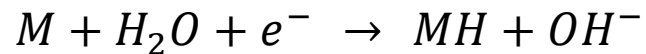
- *positive electrode:* Nickel Oxide compound
- *negative electrode:* special metal alloys (e.g. $LaNi_5$, other AB_5 composites, $TiNi_2$, ...)
- *separator:* porous polymer
- *electrolyte:* alkaline solution (based on KOH)



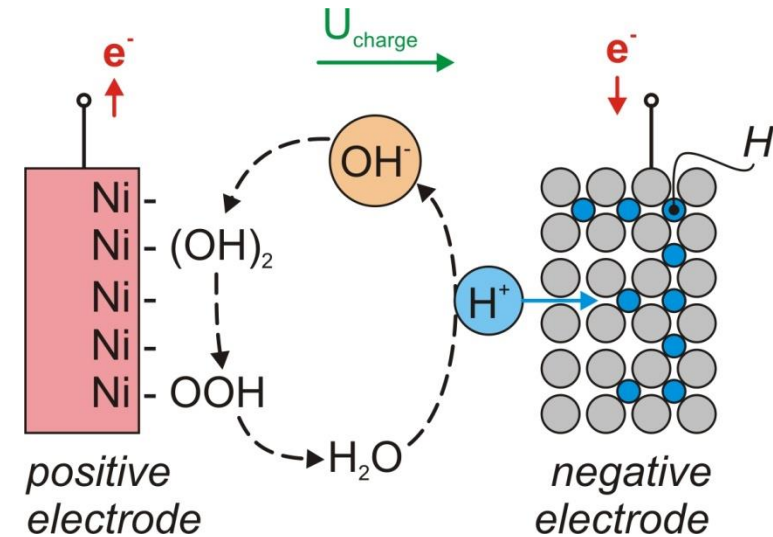
NiMH Batteries

Electrochemical Processes during Charge

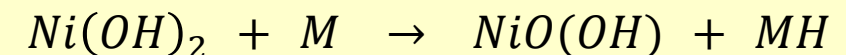
1. water electrolysis (H_2O) at the negative electrode
2. negative electrode: take-up of electrons and storage of hydrogen ($\text{H}^+ + \text{e}^-$)



3. positive electrode: use-up of OH^- and regain of H_2O via oxidation



overall charge reaction:



NiMH Batteries

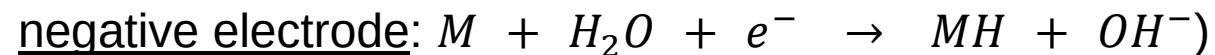
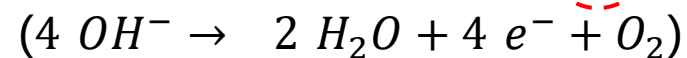
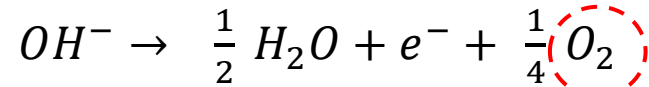
Oxygen Generation during Charge

Situation at the **end** of the charge process

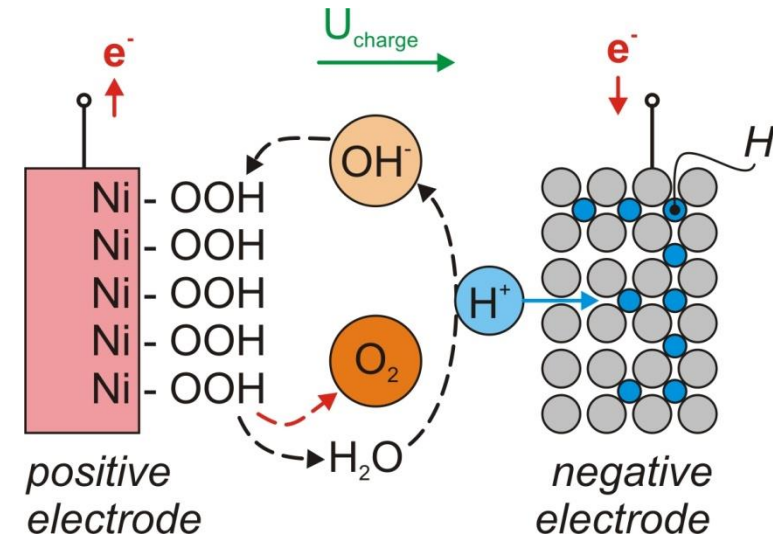
- positive electrode: fully oxidized, no OH^- storage possible
- negative electrode: partially filled with H

Consequences

- formation of O_2 gas at the positive electrode



- rise of internal cell pressure
- potential cell damage



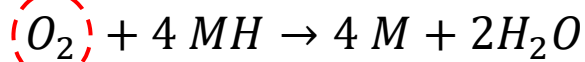
In general:

The oxidation capacity at the **positive electrode** defines the cell capacity.

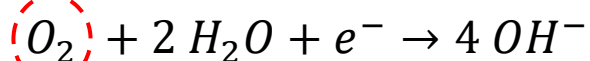
NiMH Batteries

Oxygen Consumption during Charge

1. **diffusion** of oxygen from the positive to the negative electrode
2. **diffusion** of oxygen into the negative electrode
3. formation of water with hydrogen stored in the negative electrode

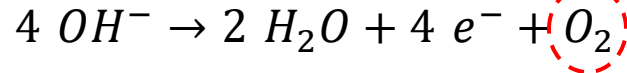
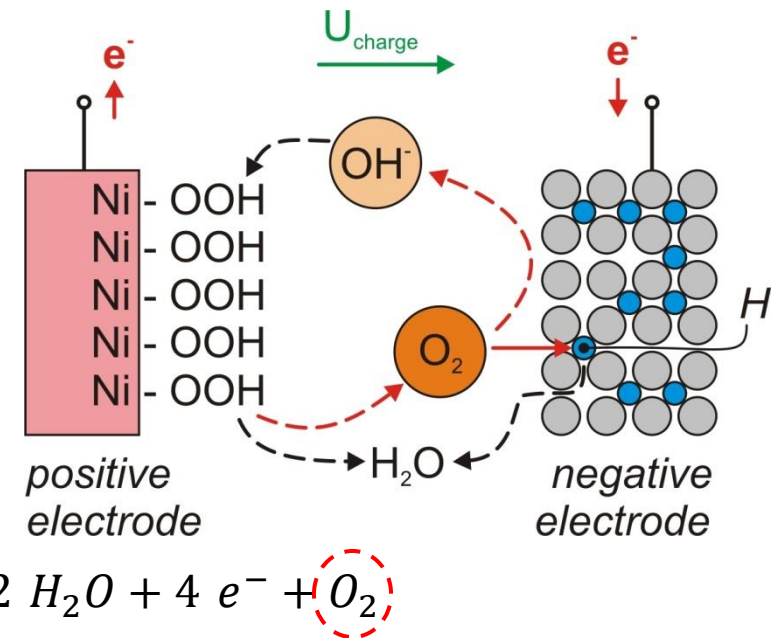


4. electrochemical formation of OH^- at the negative electrode



Consequences

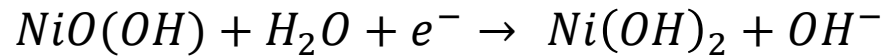
- partial use-up of stored hydrogen
- no excess formation of oxygen gas
- ➡ power dissipation instead of charging
- ➡ fully sealed cell is feasible, however ...
- ➡ charge current limitation is required (*otherwise: non-equilibrium of slow O_2 diffusion, pressure rise*)



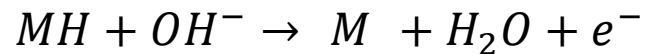
NiMH Batteries

Electrochemical Processes during Discharge

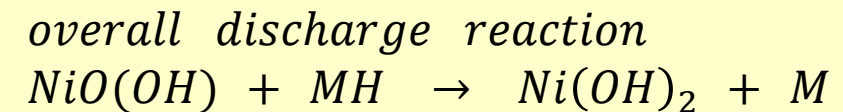
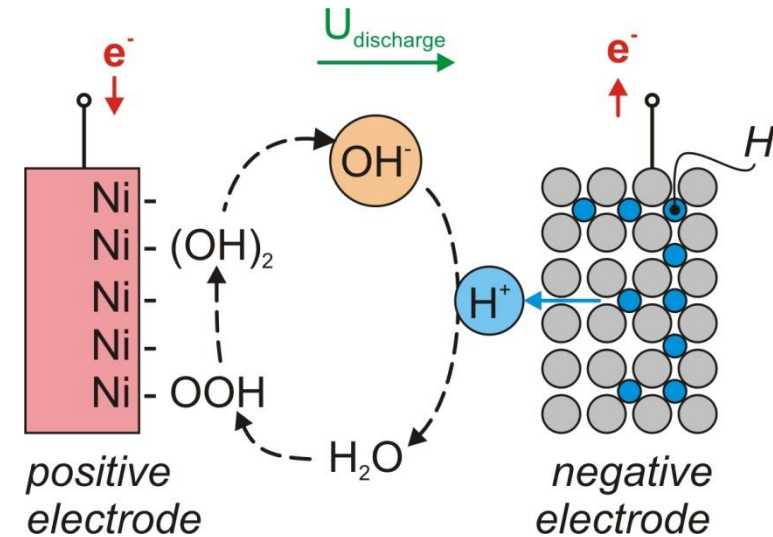
1. reduction of the positive electrode, producing OH^-



2. formation of H_2O from OH^- and H^+ taken from the negative electrode



3. release of electrons at the negative electrode



NiMH Batteries

Typical Battery Characteristics

	Value	Comments
nominal cell voltage during discharge	1.20...1.30 V (1.25 V)	depending on load current and state of charge
end-of-discharge voltage	1.0 V	
end-of-charge voltage	1.45 ... 1.55 V (1.5 V)	depends on charge current
standard charge current	0.1 x capacity C [Ah]/h	
standard charging time	14...16 hrs at $I = 0.1C$	rapid charge at higher currents possible
self-discharge (= loss of capacity)	5...30%/month	low loss cells available, influence of temperature
charge-discharge cycle lifetime	several 100	robust battery, regeneration possible after deep discharge

NiMH Batteries

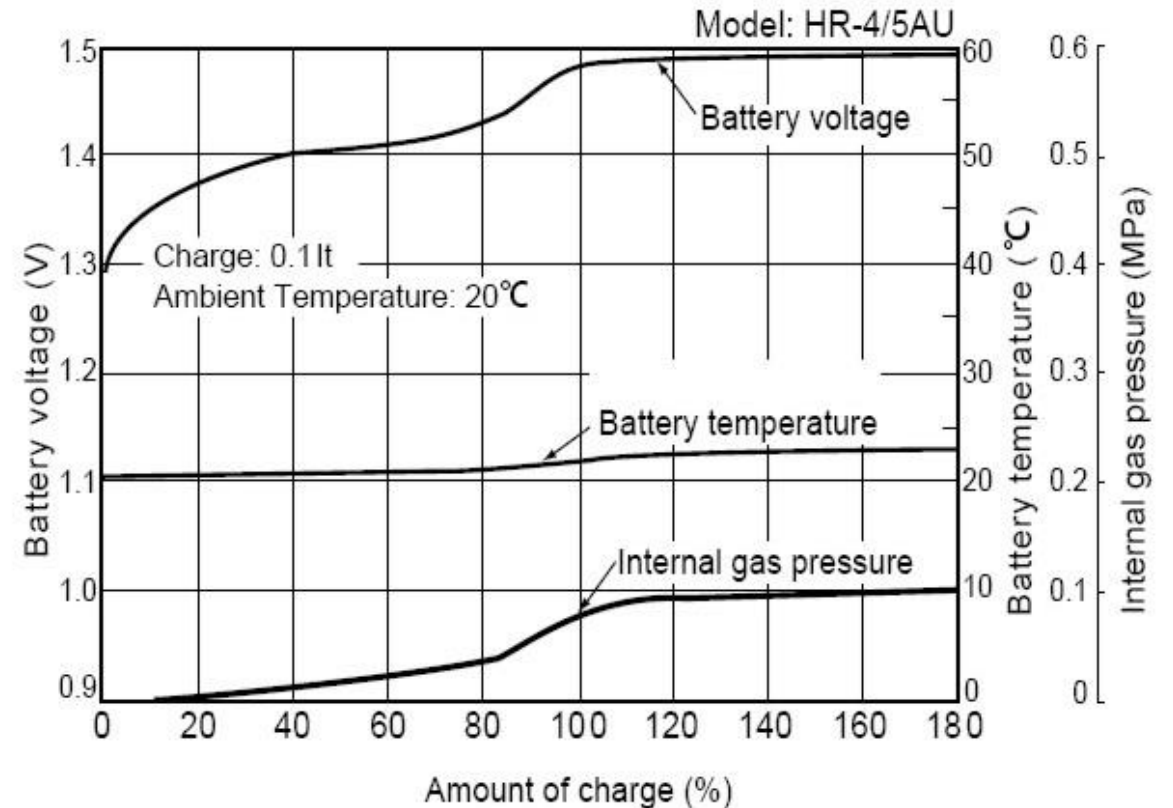
Charging

Standard charge conditions: *constant current with end-of-charge termination*

- charge with **constant current**
- determination of the end-of-charge via ...
 - ➔ time-controlled (only for fully discharged batteries)
 - ➔ detection of temperature rise
 - ➔ detection of negative voltage drop $-\Delta V$ (only reasonable for high charge currents)

Example: Sanyo Twicell HR-4/5AU
calculation of standard charge current

$$\text{rated capacity: } C = 2150 \text{ mAh}$$
$$\Rightarrow I_C = 0.1 \cdot C \cdot \frac{1}{[h]} = 215 \text{ mA}$$



NiMH Batteries

Phases during the Charging Process

1. Stabilization and start of charge: 0 ... 30%

- recovery from deep discharge
- stabilization of the electrode potentials
- on-set of H^+ generation
- higher losses due to recovery effects, etc.

2. Mid-of-Charge: 30% ... 80%

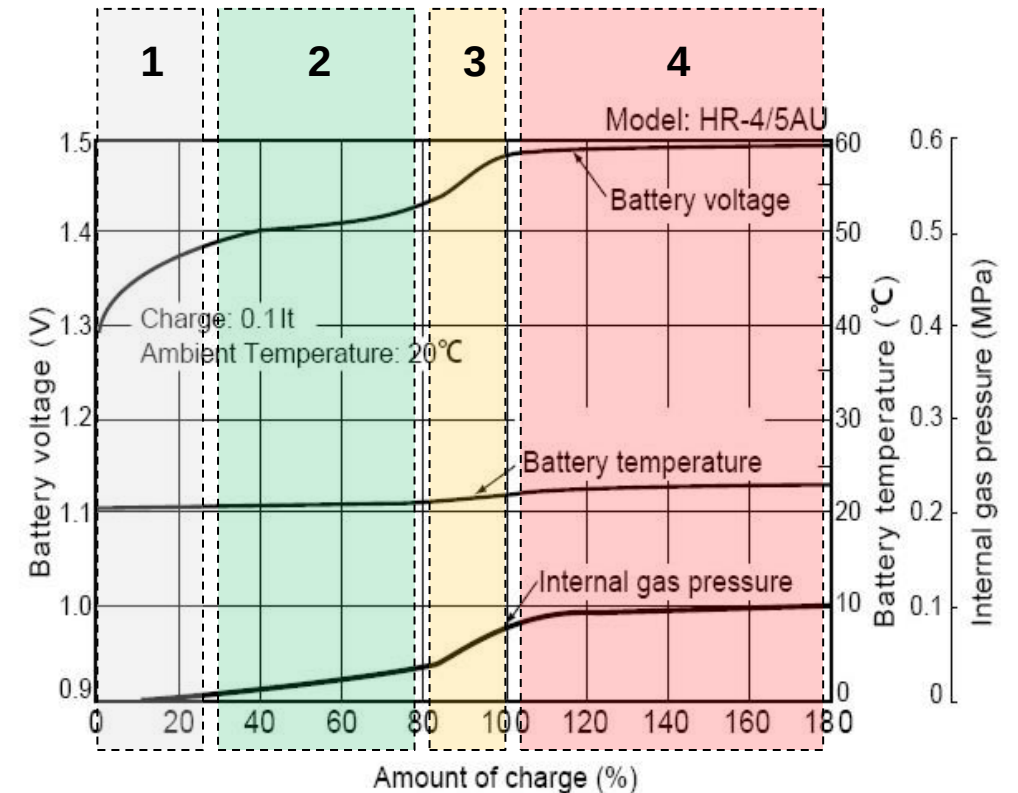
- stabilized electrode potentials
- moderate temperature and pressure
- highest charging efficiency

3. End-of-Charge: 80% ... 100%

- on-set of oxygen generation
- increased temperature and pressure
- reduction of charging efficiency

4. Over-Charge: 100% ...

- no increase of internal charge
- further increase of temperature and pressure
- high losses and potential damage

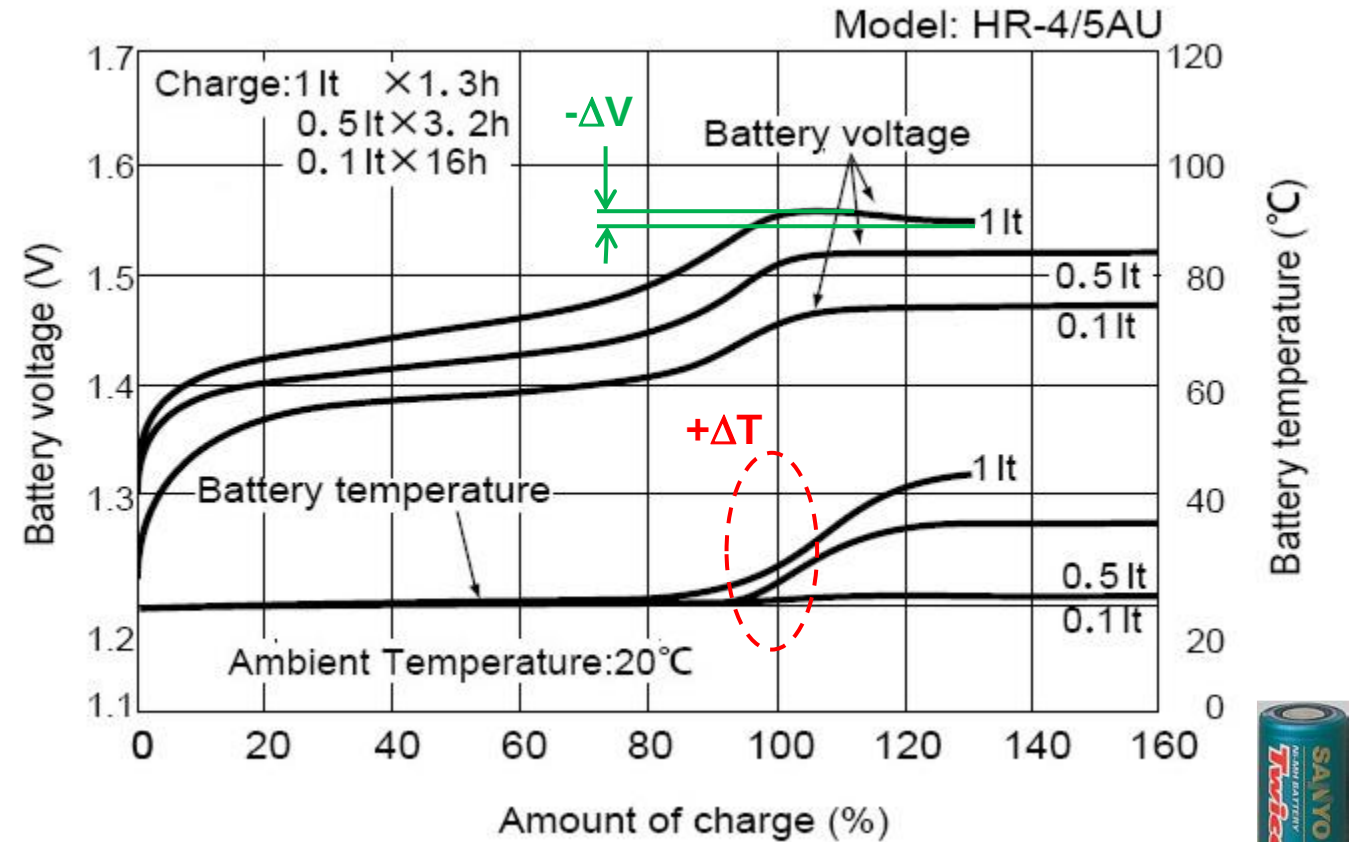


NiMH Batteries

Influence of the Charge Current

Observations at the end of charge

- rise of temperature ΔT with the charging current
frequently used for end-of-charge detection
- slight voltage drop $-\Delta V$ during overcharge (for high currents)
can be used for end-of-charge detection



Sanyo Twicell HR-4/5AU

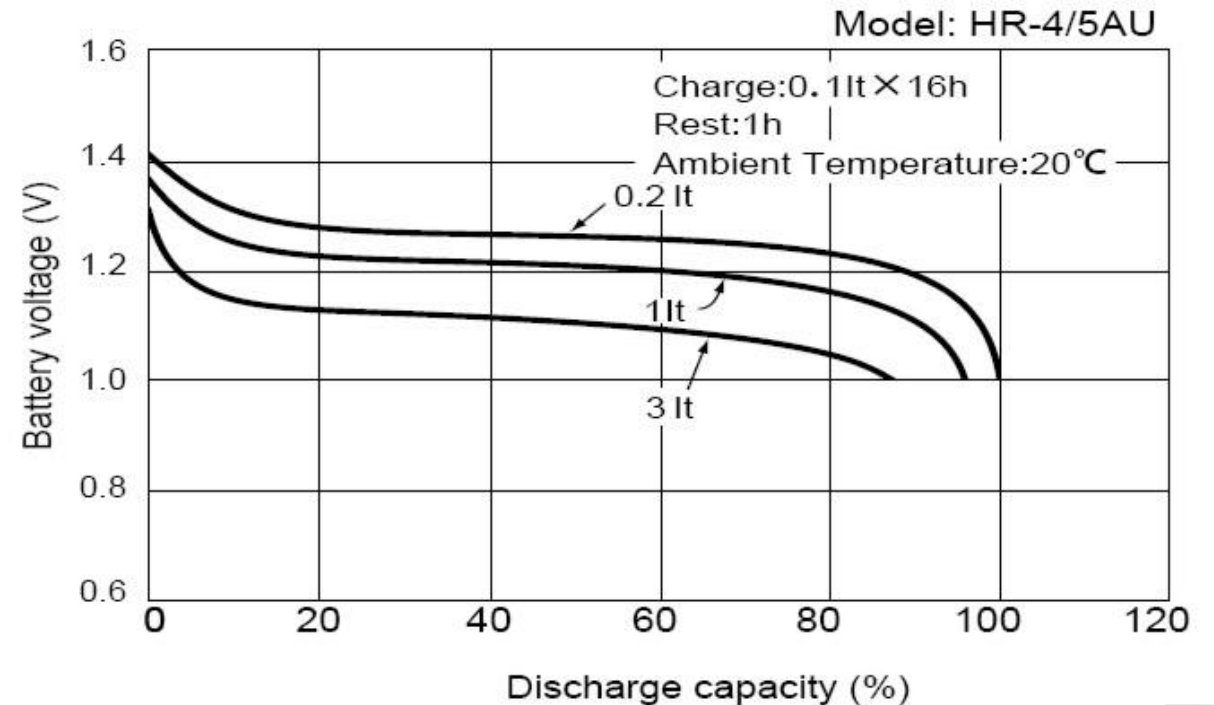


NiMH Batteries

Discharge Characteristics

Observations

- relatively stable voltage plateau
nominal value: $\sim 1.25\text{ V}$
- cell voltage depends on discharge current
- steep fall of cell voltage at the end of discharge
typical end-of-discharge voltage: 1.0 V

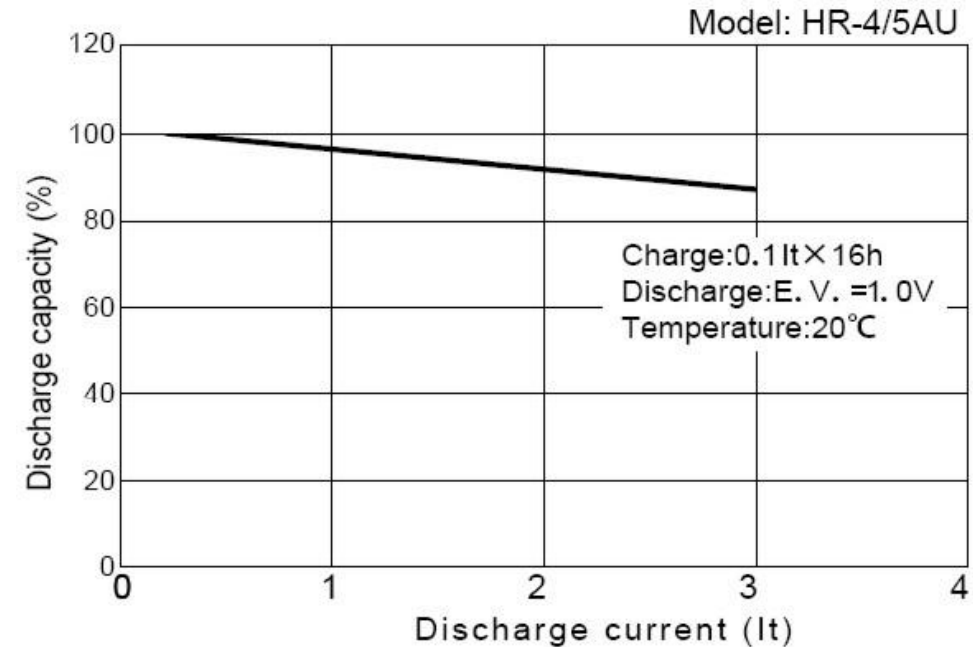
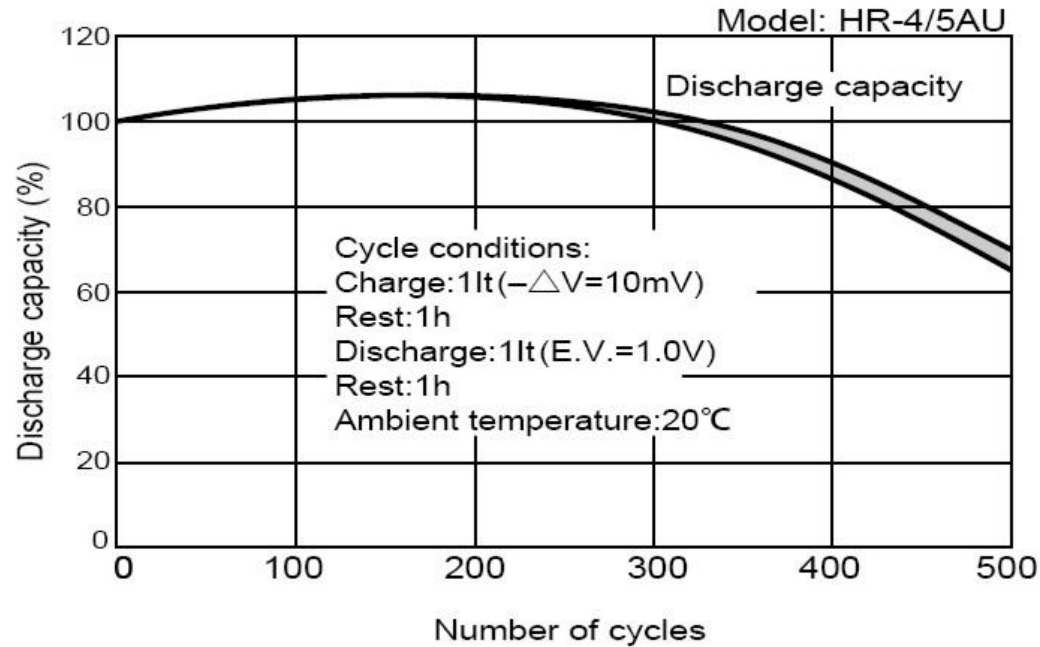


Sanyo Twicell HR-4/5AU



NiMH Batteries

Cycle Life and Discharge Current



Observations

- high cycle lifetime /several 100 cycles)
- loss of capacity with higher discharge currents

Sanyo Twicell HR-4/5AU



NiMH Batteries

Coulomb and Energy Efficiency

Example: Sanyo Twicell HR-4/5AU

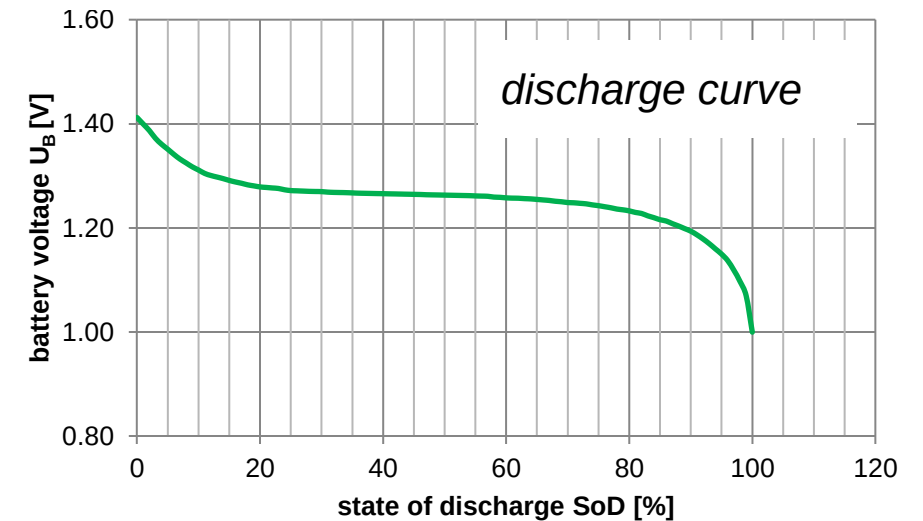
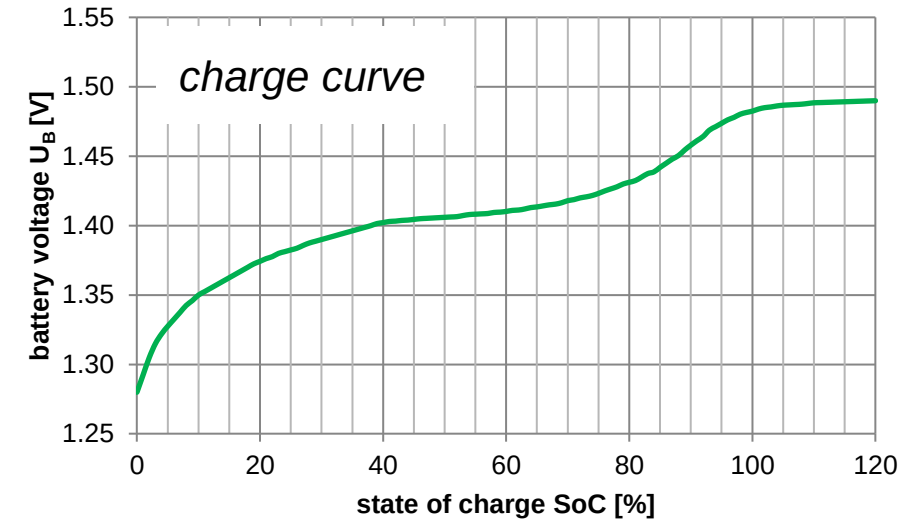
- *nominal capacity:* 2150 mAh
- *charging conditions:* 16 h at 215 mA (0.1C)
- *discharge conditions:* 5 h at 430 mA (0.2C)

Coulomb efficiency

$$CE = \frac{I_d \cdot T_d}{I_c \cdot T_c} = \frac{430 \text{ mA} \cdot 5 \text{ h}}{215 \text{ mA} \cdot 16 \text{ h}} = \frac{7740 \text{ As}}{12384 \text{ As}} = 62.5 \%$$

Energy efficiency

$$EE = \frac{\int_{t=0}^{T_d} U_B \cdot I_d dt}{\int_{t=0}^{T_c} U_B \cdot I_c dt} = \frac{2.722 \text{ V} \cdot \text{A} \cdot \text{h}}{4.877 \text{ V} \cdot \text{A} \cdot \text{h}} = 55.8 \%$$



NiMH Batteries

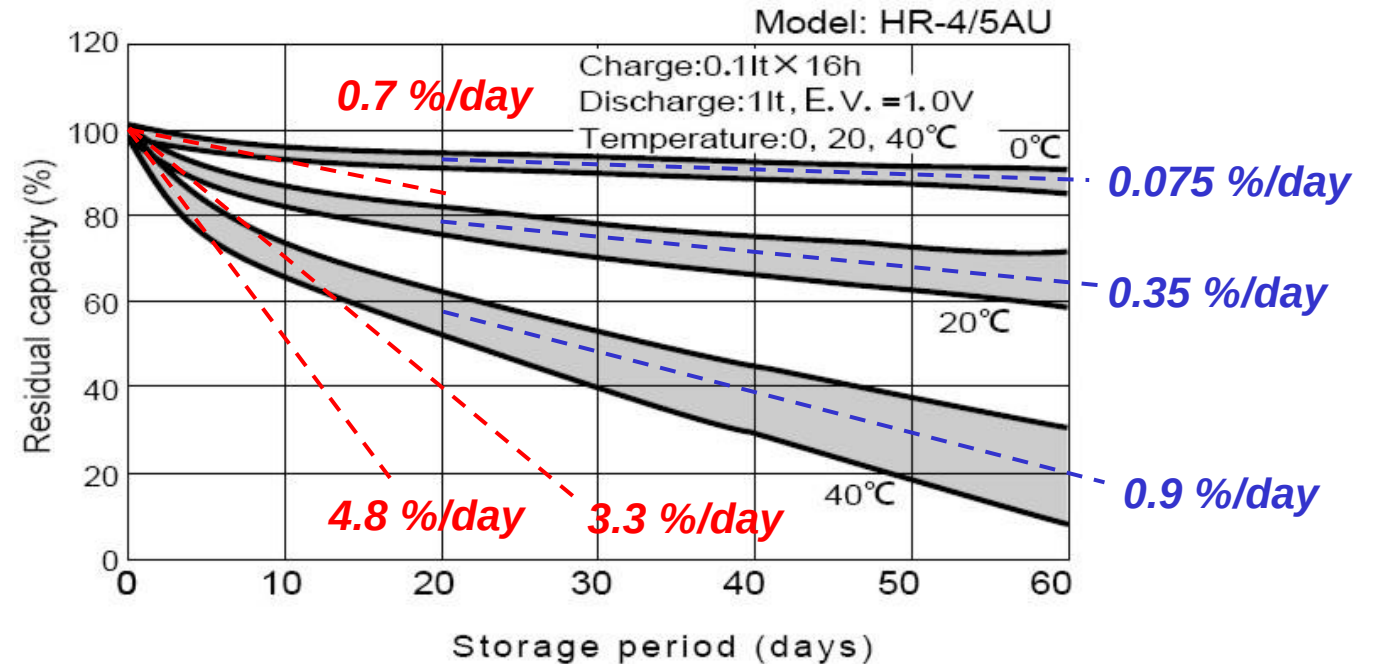
Self-discharge Rate SDR

- ... gives the relative loss of capacity over a certain period of time

$$SDR = \frac{\Delta C / C_0}{\Delta t}$$

Observations

- higher, nonlinear loss of capacity during the first days (fully charged battery)
- linear loss of capacity during long-term storage (> 1 month)
- higher losses at elevated temperature



Sanyo Twicell HR-4/5AU



NiMH Batteries

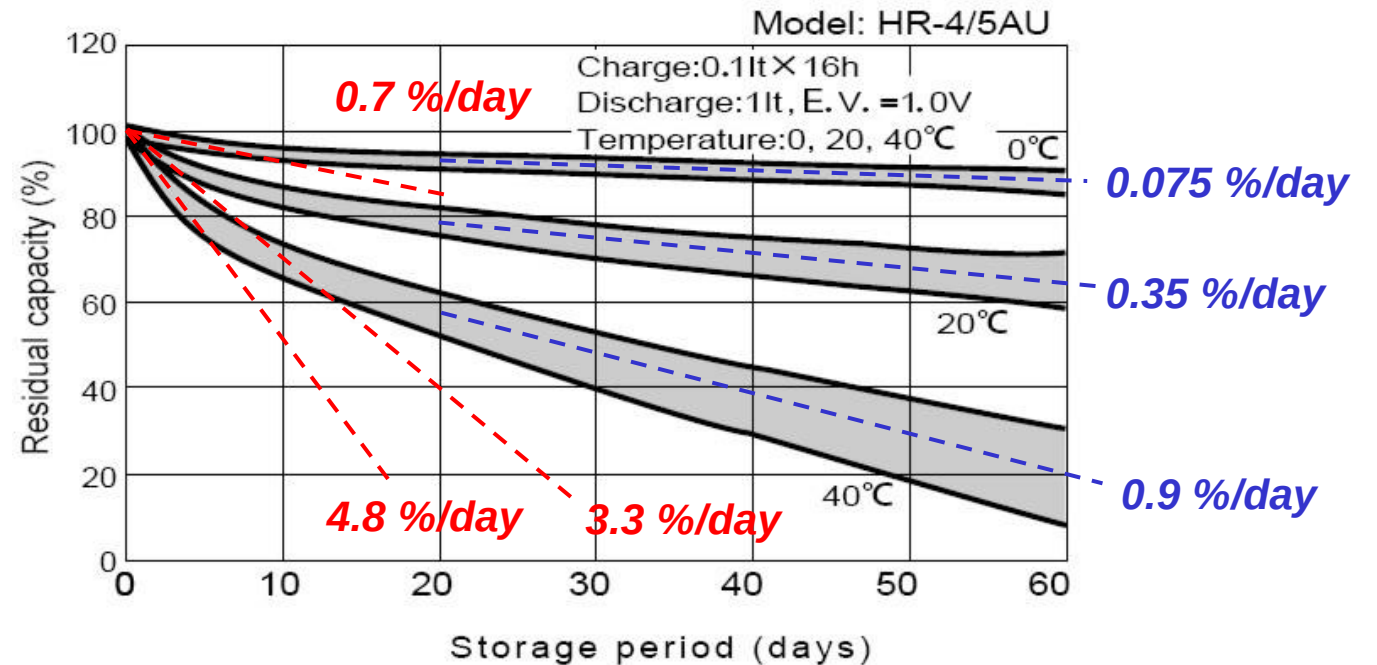
Effective Leakage Current

$$I_L = SDR \left[\frac{1}{\Delta t[h]} \right] \cdot C[Ah]$$

Example: Sanyo Twicell HR-4/5AU

$$\left. \begin{array}{l} T = 20^\circ\text{C} \\ C_0 = 2150 \text{ mAh} \\ \text{initial: } SDR = 3.3 \text{ \%/day} \\ \text{longterm: } SDR = 0.35 \text{ \%/day} \end{array} \right\} \Rightarrow$$

$$\left\{ \begin{array}{l} \text{initial: } I_L = 2.96 \text{ mA} \\ \text{longterm: } I_L = 0.31 \text{ mA} \end{array} \right.$$



Sanyo Twicell HR-4/5AU



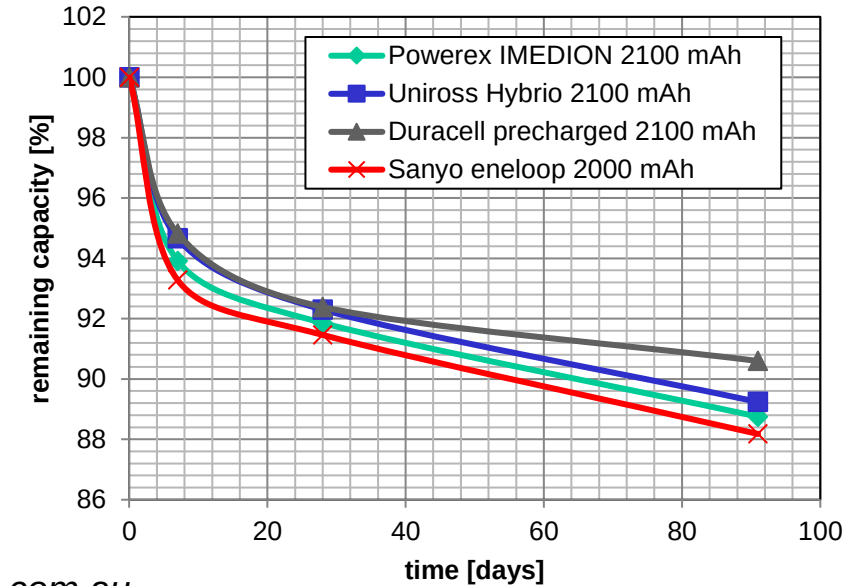
NiMH Batteries

Low Self-discharge Batteries

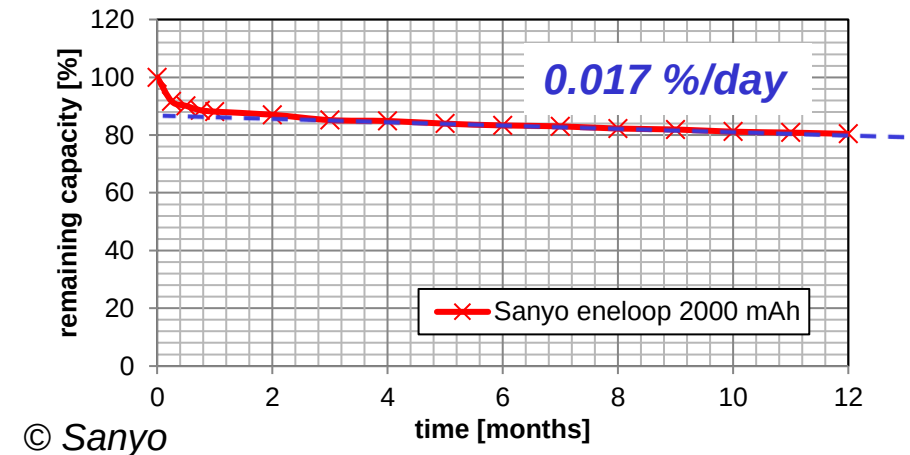
- standard NiMH chemistry, however with optimized and higher purity materials
- customized for low leakage effects during long storage

Available brands (back in 2011)

- Sanyo: Eneloop today: Panasonic
- Duracell: Pre Charged / Duraloops
- Powerex: IMEDION
- Rayovac: Hybrid today: Varta
- Uniross: Hybro
-



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NiMH Batteries

Lifetime

Non-optimal conditions for a long lifetime

- deep discharge ➡ depolarization of the electrodes
- overcharge ➡ rise of temperature, pressure, mechanical stress
- temperature excesses ➡ danger of freezing or leakage,
higher self-discharge rate at elevated temperatures

Optimal conditions for a long lifetime

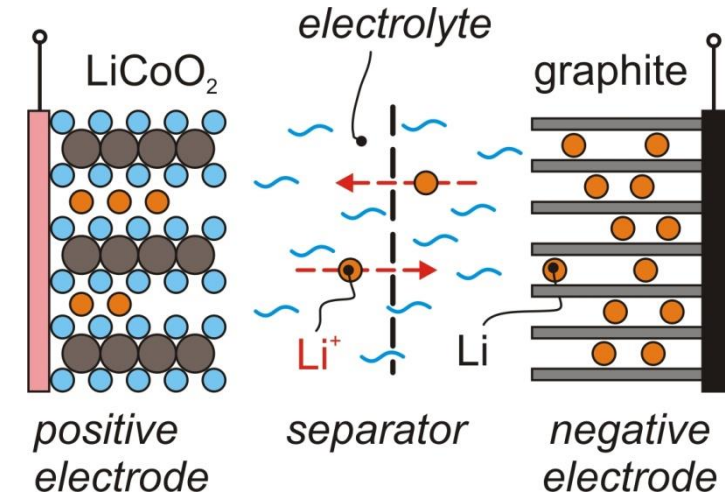
- shallow charge-discharge between 40% and 60% ➡ ideal battery chemistry, highest charging efficiency, stable voltage, low self-discharge rate
- small to average charge currents (typ. 0.1C ... 1 C) ➡ ideal battery chemistry, high Faraday efficiency, acceptable temperature rise



Lithium-Ion (Li-Ion) Batteries

Set-up and Working Principle

- transport of charged Li ions between anode and cathode („rocking chair battery“)
- external current via neutralization of the ion charges transported to/from the electrodes



- *positive electrode:* lithium metal oxide (e.g. LiCoO_2)
- *negative electrode:* special graphite
- *separator:* porous polymer
- *electrolyte:* water-free organic electrolyte (salt e.g. LiPF_6 dissolved in e.g. ethylene carbonate)

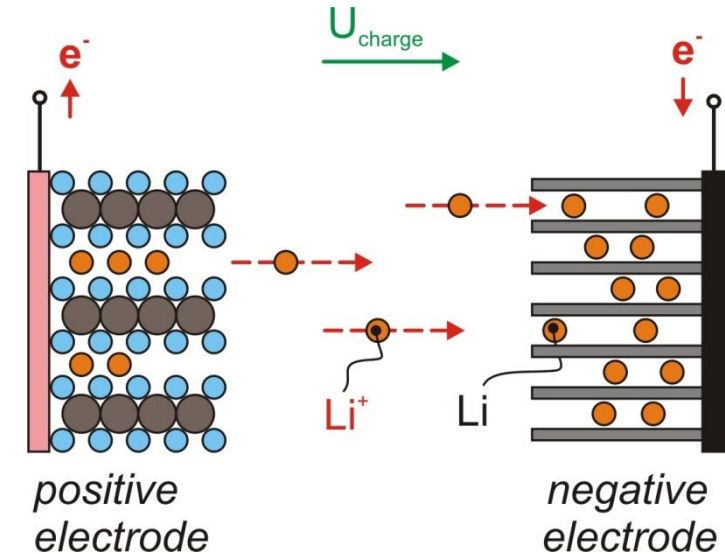
Li-Ion Batteries

Electrochemical Processes during Charge

1. Electrons are withdrawn from the positive electrode
 ➔ *Li at the positive electrode is converted into Li^+ ions*

$$\text{LiCoO}_2 \rightarrow \text{Li}_{1-x}\text{CoO}_2 + x \cdot e^- + x \cdot \text{Li}^+$$
2. Li^+ ions migrate to the negative electrode and are intercalated into the graphite layer
3. Neutralization of the positive ion charge with electrons from the negative electrode
 ➔ *Li^+ ions at the negative electrode are converted into Li atoms*

$$\text{C}_6 + x \cdot e^- + x \cdot \text{Li}^+ \rightarrow \text{Li}_x\text{C}_6$$



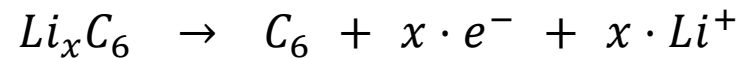
overall charge reaction:

$$\text{LiCoO}_2 + \text{C}_6 \rightarrow \text{Li}_{1-x}\text{CoO}_2 + \text{Li}_x\text{C}_6$$

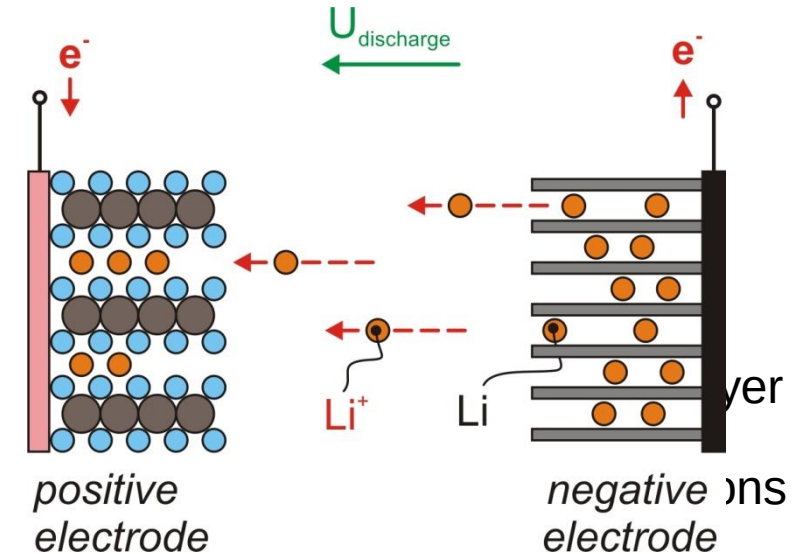
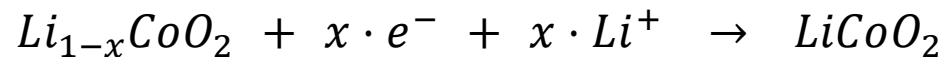
Li-Ion Batteries

Electrochemical Processes during Discharge

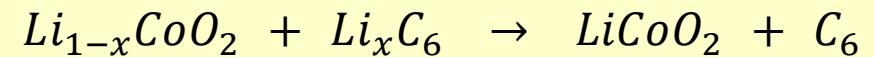
1. Electrons are withdrawn from the negative electrode
➡ *Li from the negative electrode is converted into Li^+ ions*



2. Li^+ ions migrate to the positive electrode
3. During this reaction the positive ion charge from the positive electrode
➡ *Li^+ ions travelling to the positive electrode are converted into Li atoms fixed in the $\text{Li}_{1-x}\text{CoO}_2$ layer*



overall discharge reaction:



Li-Ion Batteries

Typical Battery Characteristics

	Value	Comments
nominal cell voltage during discharge	3.3 ... 3.7 V	depending on the choice of anode and cathode materials
end-of-discharge voltage	2.03.0 V	depending on chemistry, deeper discharge will damage the battery
end-of-charge voltage	4.0 ... 4.2 V	depending on the choice of anode and cathode materials
standard charge current	0.1...1.0 x C [Ah]/h	
standard charging time	directly related to capacity C	rapid charge at higher currents possible
self-discharge (= loss of capacity)	0.5...3%/month	influence of temperature
charge-discharge cycle lifetime	several 100	ageing process during every cycle

Li-Ion Batteries

Comparison to NiMH and other Batteries

Advantages

- No memory effect
- accepts charge and discharge current in all states (discharged, partially charged)
- reasonable cell voltage ($\sim 3\text{V}$)
- acceptable self-discharge rate ($\sim 3\%/ \text{month}$)
- prolonged lifetime, if not fully charged/discharged
- high energy density
- small size and weight
- low toxicity

Disadvantages

- reduced lifetime, when operated at elevated temperatures
- reduced capacity at low temperatures (like all batteries)
- limited acceptance of trickle charge
- more sophisticated power management required

Li-Ion Batteries

Standard Charge Conditions

Usually: *Constant current – constant voltage (CCCV)*

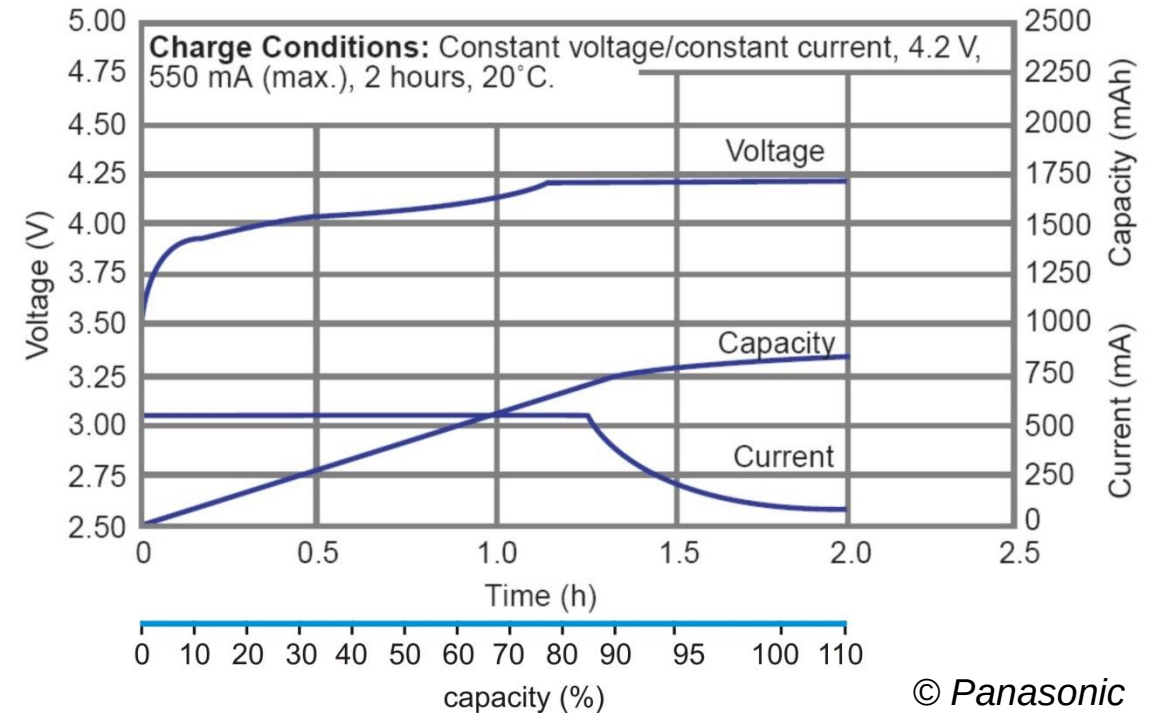
- charge with **constant current**: $I_C = 0.1 \dots 1 C$ (maximum value is given in the datasheet)
- Charge voltage **limitation** to a fixed end-of-charge voltage: $U_{max} = 4.2 \text{ V/cell}$
(only slight variations, depending on the battery type)

Example: Panasonic CGR 17500

calculation of standard charge current

rated capacity: $C = 830 \text{ mAh}$

datasheet: $I_C = 0.66 \cdot C \cdot \frac{1}{[h]} = 550 \text{ mA}$



Li-Ion Batteries

Safety Precautions during Charge

Charge control electronics for ...

- reverse polarity protection
 - charge temperature monitoring: charge only for $0^{\circ}\text{C} < T < 45^{\circ}\text{C}$
 - Charge current limit, e.g.: $I_C < 0.7\text{ C}$
 - Discharge current protection: detection of internal short circuits and current limit
 - Charge voltage limit: break of a fuse, when overvoltage is applied (danger of explosion)
 - Overcharge protection: stop of charging for $U_B > 4.3\text{ V}$
 - Overdischarge protection: stop of discharge for $U_B < 2.3\text{ V}$ (varies with manufacturer)
 - Temperature fuse: opens for $T > 100^{\circ}\text{C}$ (danger of explosion)
- ➡ (often) IC integrated function combined with e.g. MPPT for harvesters, input buck-boost-converters, ...

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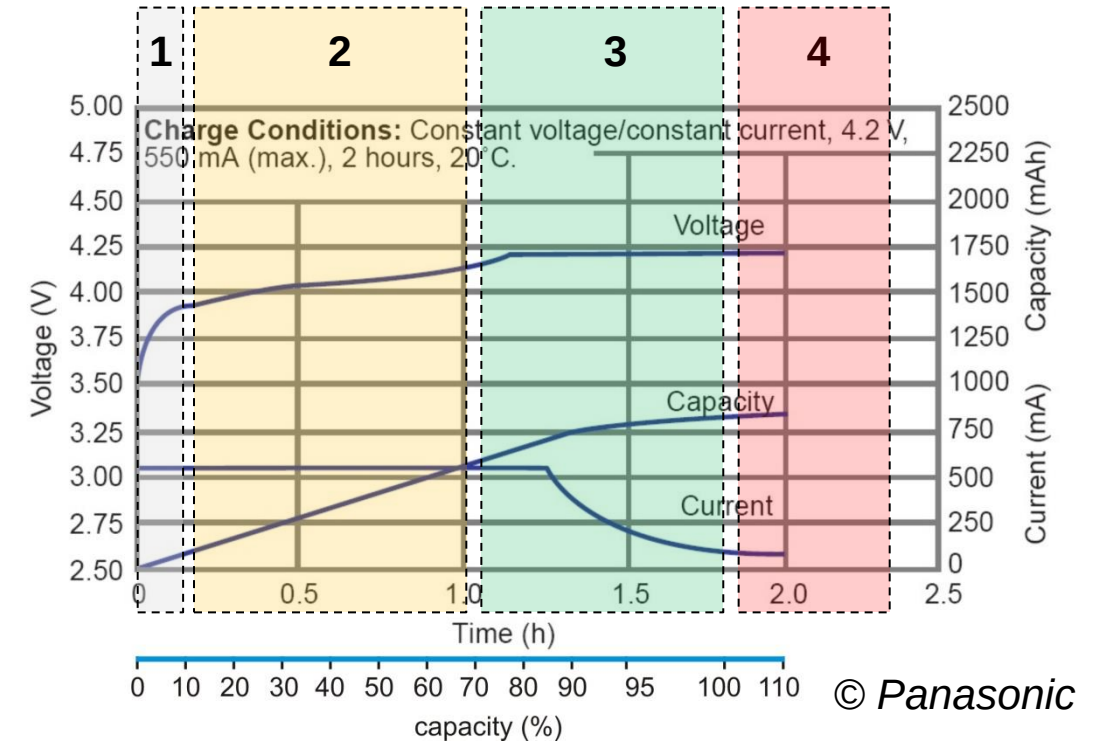
data taken from: www.powerstream.com

Li-Ion Batteries

Phases during the Charging Process

1. Stabilization and start of charge: 0 ... 10%

- stabilization of the electrode potentials
- on-set of Li^+ ion transfer
- higher losses due to recovery effects, etc.
- **2. Mid-of-Charge: 10% ... 70%**
 - stabilized electrode potentials
 - continuous transfer of Li^+ ions
 - highest charging efficiency
- **3. End-of-Charge: 80% ... 100%**
 - voltage is fully limited by charge electronics
 - charge current drops = lower transfer of Li^+ ions
 - on-set of losses = lower charging efficiency



4. Over-Charge: 100% ...

- no increase of internal charge = all ions transferred
- increase of temperature and pressure
- high losses, ageing effects and potential damage

Li-Ion Batteries

Charge Acceptance Rate

- acceptance of charge and discharge in **every** state-of-charge, however ...
- **different charging rates**, depending on the ...
 - ➔ actual cell voltage and state of charge
 - ➔ externally applied voltage

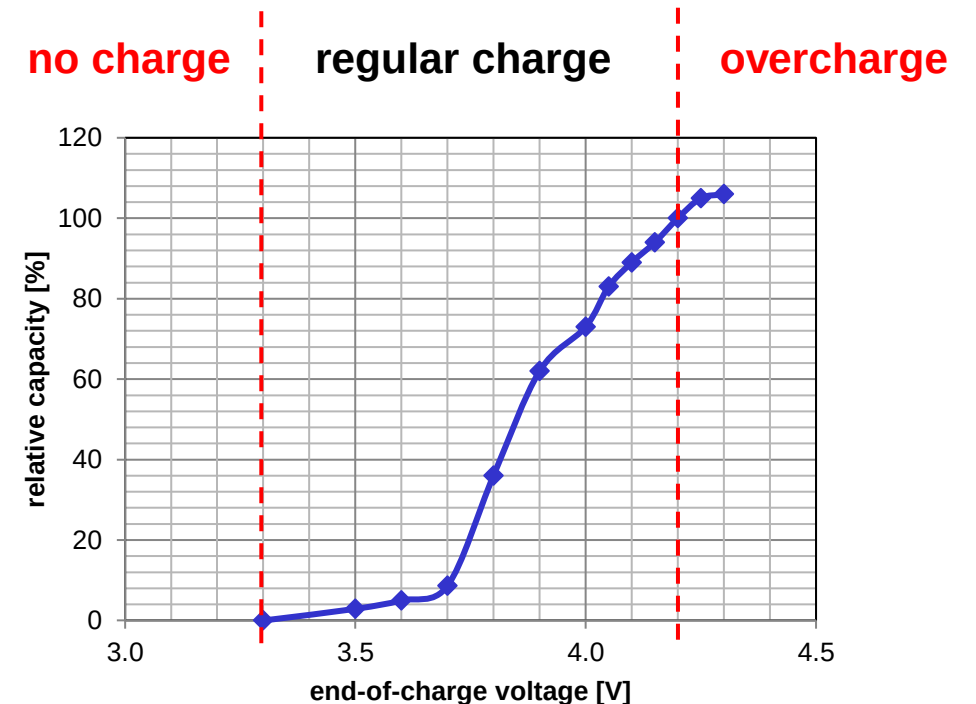
Example: Powerstream PGEB0054253

rated capacitance: 60 mAh

rated end-of-charge voltage: 4.2 V

procedure: discharged cell is charged
with different end-of-charge voltages

- no acceptance of charge for voltages below 3.3 V
- highest acceptance rate for voltages between 3.7 V and 4.0 V
- saturation and potential cell damage above 4.2 V



data taken from: www.powerstream.com

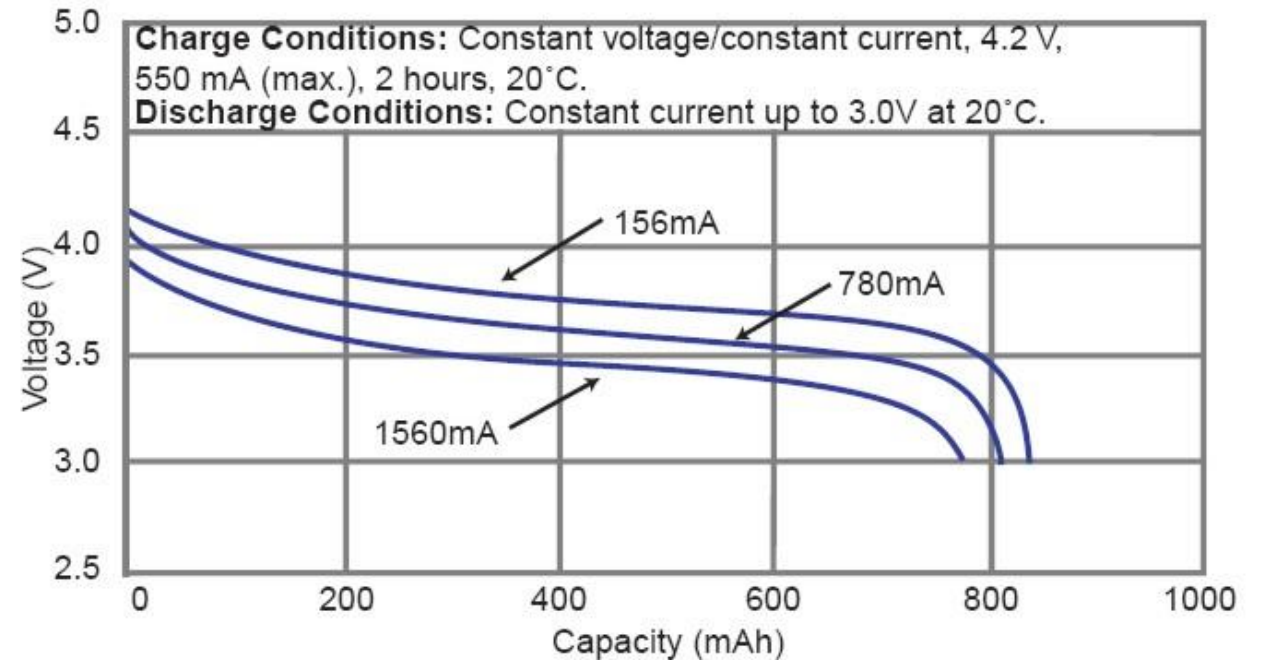
Li-Ion Batteries

Discharge Characteristics



Observations

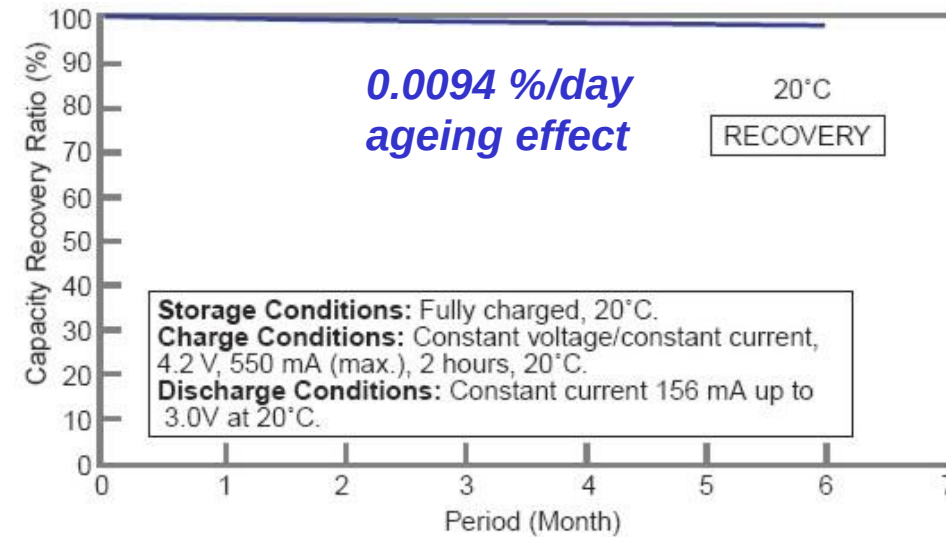
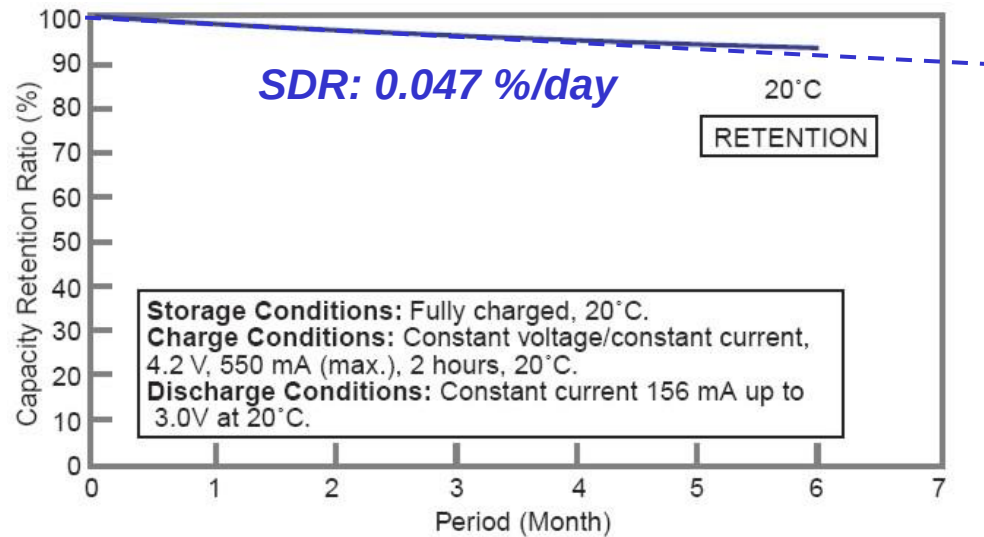
- relatively stable voltage plateau
nominal value: $\sim 3.6\text{ V}$
- cell voltage depends on discharge current
- steep fall of cell voltage at the end of discharge
typical end-of-discharge voltage: 3.0 V



Panasonic CGR 17500

Li-Ion Batteries

Self-discharge Rate and Ageing under Storage Conditions



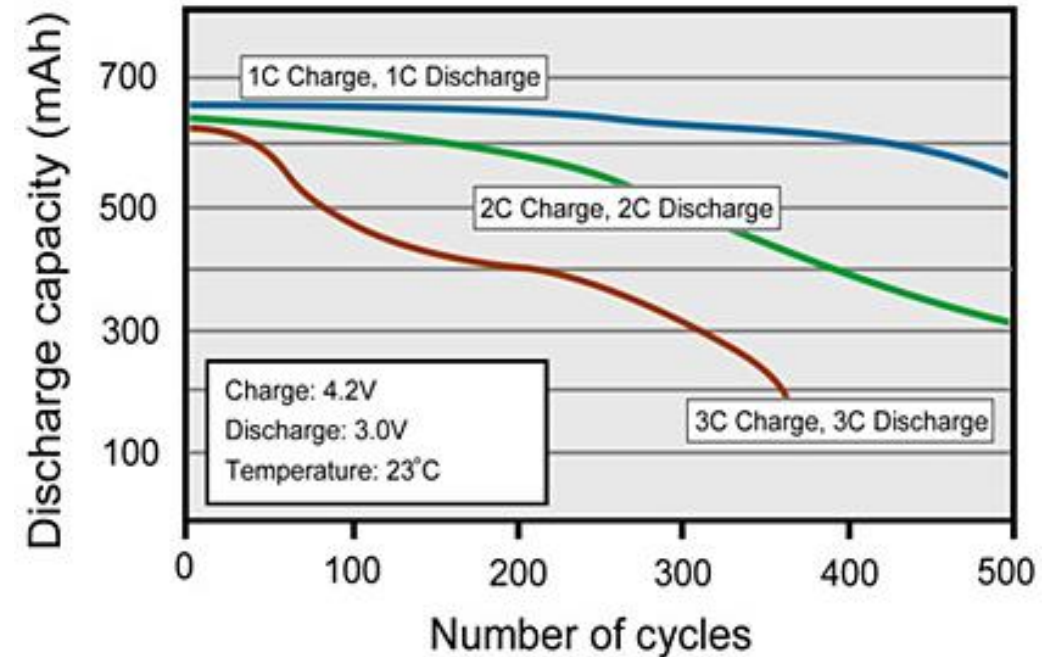
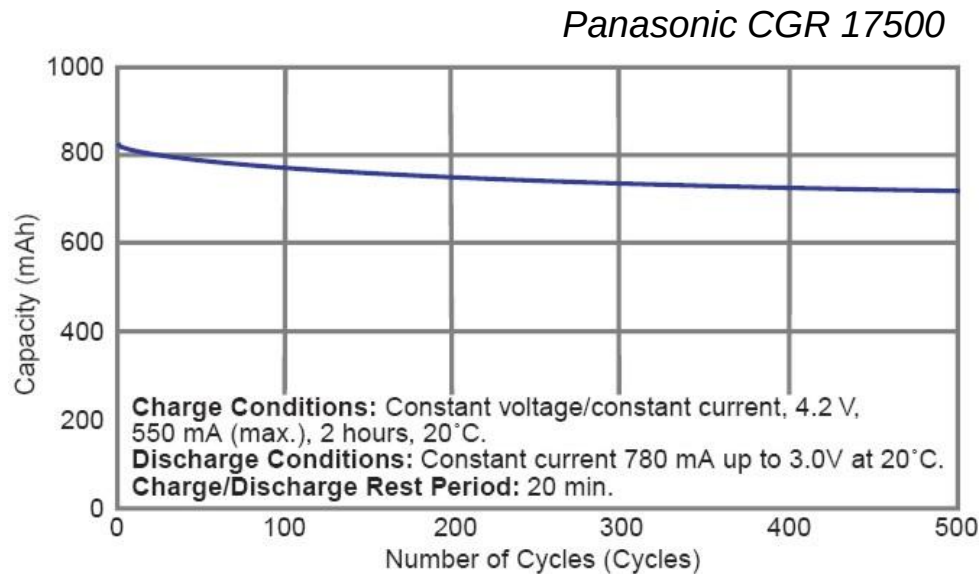
Panasonic CGR 17500

Observations

- slightly higher, nonlinear loss of capacity during the first days (fully charged battery)
- in general, almost linear loss of capacity during long-term storage (> 1 month)
- continuous loss of capacity (**ageing** effect)
- not shown here: higher losses and ageing effects at higher temperatures

Li-Ion Batteries

Cycle Life and Influence of the Charge/Discharge Current



Observations

- instantaneous on-set of ageing
- loss of cycle lifetime with high charge/discharge currents
- high cycle lifetime (several 100 cycles) under appropriate conditions (0.8 C recommended)

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Li-Ion Micro-batteries

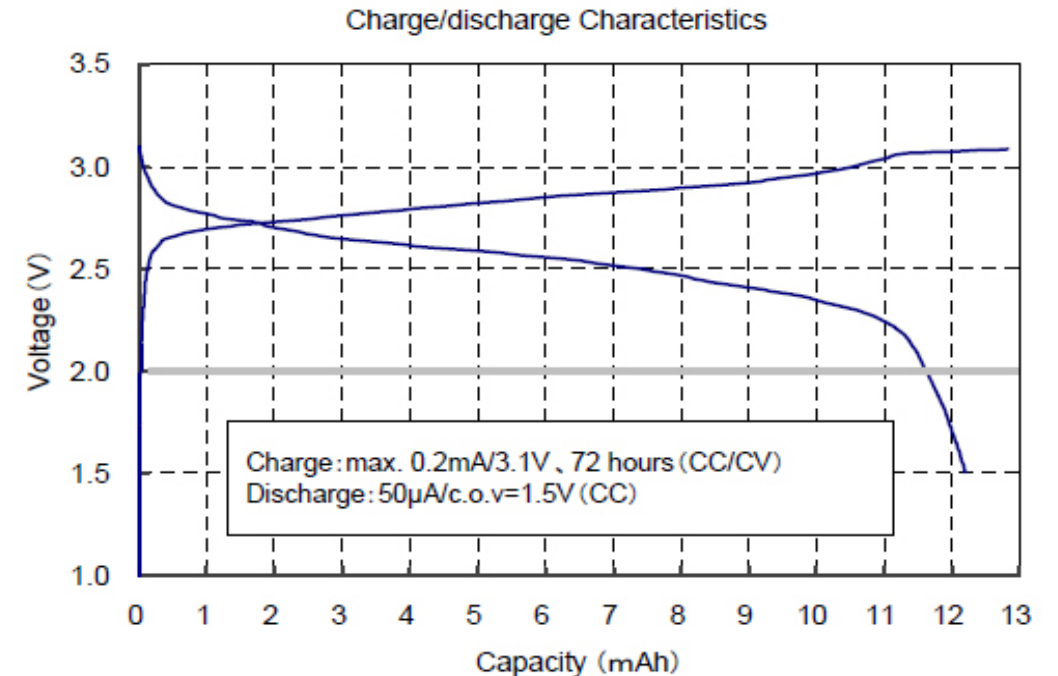
Typical Characteristics of a Button Cell

Example: Seiko MS 920 SE 3V/11mAh

Nominal voltage	3.0 V
Charge voltage range	2.8 ... 3.3 V
Standard charge voltage	3.1 V
Nominal capacity C	11 mAh
Standard charge conditions	max. 0.2 mA @ 3.1 V for 72 hrs
Standard discharge current	50 μ A
End-of-discharge voltage	2.0 V
Cycle life (no. of cycles)	
100% discharge per cycle	100
20% discharge per cycle	1000



Ø 9.5 mm x 2.8 mm,
weight 0.47 g



Li-Ion Micro-batteries

Brown-out Capabilities and Lifetime

Seiko MS 920 SE

Setup: A low-power sensor system runs on 50 μW at min. 2.3 V

$$I_L = \frac{50 \mu W}{2.3 V} = 22 \mu A \Rightarrow R_L = \frac{2.3 V}{22 \mu A} = 105 k\Omega$$

1. Discharge over the full capacity (100 cycles):

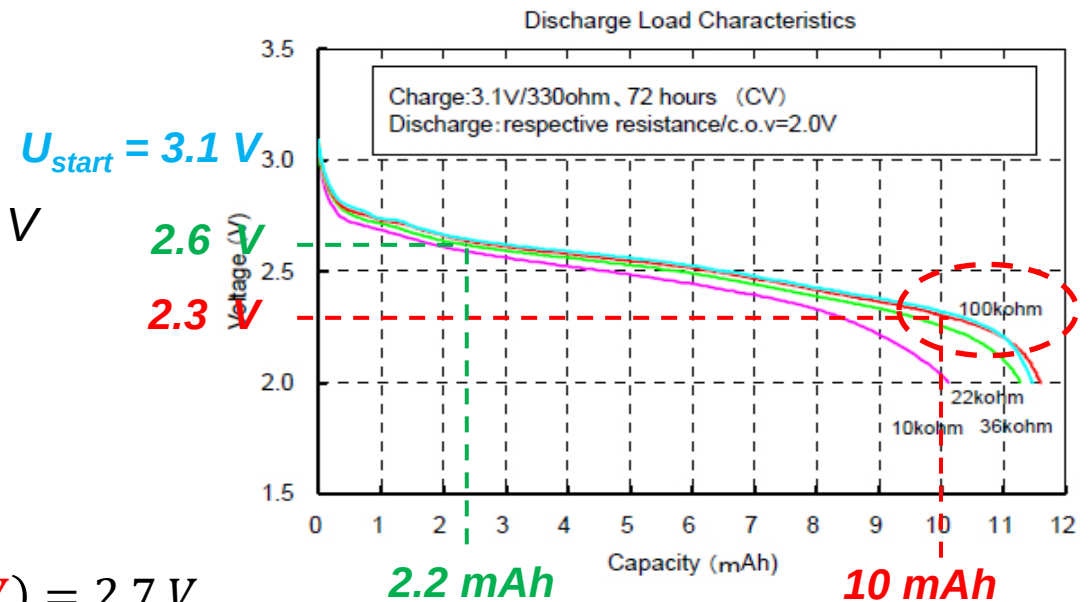
$$\text{average voltage: } U_{av} = \frac{1}{2} (U_{start} + U_{end}) = \frac{1}{2} (3.1 V + 2.3 V) = 2.7 V$$

$$\text{capacity: } C_{100} = 10 mAh = 0.91 \cdot C_{max} \Rightarrow t_{100} = \frac{C_{100}}{U_{av}} \cdot R_L = 389 h \Rightarrow t_{Life} = 100 \cdot T_{100} = 4.4 \text{ years}$$

2. Discharge to 20 % of C_{max} (1000 cycles):

$$\text{average voltage: } U_{av} = \frac{1}{2} (U_{start} + U_{end}) = \frac{1}{2} (3.1 V + 2.6 V) = 2.85 V$$

$$\text{capacity: } C_{1000} = 2.2 mAh = 0.2 \cdot C_{max} \Rightarrow t_{1000} = \frac{C_{1000}}{U_{av}} \cdot R_L = 81 h \Rightarrow t_{Life} = 1000 \cdot T_{1000} = 9.3 \text{ years}$$



Rechargeable Batteries

Conclusion

- Can offer medium to high Coulomb and Energy Efficiency
- Some have very low self-discharge (leakage current)
- Capacity depends on output current
- Lifetime can be improved if only partially charged around optimum working point
- Temperature dependence (especially lithium ions)
- Dimension according to
 - Average voltage between full and empty
 - Capacity needed
 - Cycle life
 - Weight and size